

UC-02 – Reserve Parking Bay

Problem Statement #22 – Smart Parking Space Rental & License Validator

Primary Actor	Vehicle Owner
Supporting Actor	Smart Parking System
Preconditions	The parking facility is operational; parking bays and their real-time availability are registered; the vehicle owner has a registered vehicle with a valid license plate number.
Postconditions	The selected parking bay is reserved; the reservation is linked to the vehicle license plate; reservation start time and permitted duration are recorded; confirmation is shown to the vehicle owner.

Main Success Scenario

1. Vehicle Owner opens the Smart Parking system.
2. System displays available parking facilities and bays in real time.
3. Vehicle Owner selects a parking facility and preferred parking time.
4. System displays currently available parking bays.
5. Vehicle Owner selects an available parking bay.
6. Vehicle Owner enters or selects the vehicle license plate number.
7. System validates the reservation details.
8. System confirms that the selected bay is still available.
9. System creates the parking reservation.
10. System records the reservation start time and permitted parking duration.
11. System changes the selected bay status to Reserved.
12. System displays the reservation confirmation to the Vehicle Owner.

Alternate Flow – Selected Bay No Longer Available

- 8a. System determines that the selected bay has already been reserved by another vehicle.
 - 8a1. System displays "Parking Bay No Longer Available".
 - 8a2. System displays the currently available bays.
 - 8a3. Vehicle Owner selects another available bay.
 - 8a4. System continues from Step 6.
 - 8a5. If no suitable bay is available, the reservation process is cancelled.